HACKCULT '26 | IIT GANDHINAGAR × NXTWAVE • PROBLEM HC26-CULT01

DHAROHAR 4D (धरोहर 4D)

4D Spatiotemporal Digital Twin & Cultural Heritage Authenticity Engine

Preserving Ancient Monuments through Multimodal Archiving, WebXR AR, and Cryptographic Authenticity

Team: Believers (Team Leader: Prachi Zala & Yash Patadiya) | **Live: https://dharohar-4d.onrender.com**

THE CHALLENGE

The Cultural Heritage Crisis: Preserving What Is Being Lost

Why 3,600+ monuments need an AI-driven, authentic 4th-dimensional digital twin today.

CONSERVATION BARRIER

1. Physical Degradation

- Over 3,690 centrally protected ASI monuments are subjected to weathering, urban expansion, and environmental erosion.
- Physical restoration is irreversible if original geometric baselines and architectural states are lost.
- Traditional laser LIDAR scanning is prohibitively expensive and cannot capture historical epochs.

DATA DISCONNECTION

2. Archival Fragmentation

- Critical historical records are trapped in 19th-century colonial surveys (James Burgess, Cousens) and scattered archives.
- Tourist photos, academic field notes, and local epigraphs remain siloed in incompatible formats.
- Zero standardized metadata schema linking physical artifacts to architectural epochs.

AUTHENTICITY FAILURE

3. Hallucination Risk

- Generic GenAI tools invent fictional carvings, wrong dynasties, and fabricated architectural timelines.
- Lack of strict ground-truth verification undermines scholarly credibility.
- Museums and scholars cannot trust unverified digital reconstructions without provenance.

OUR INNOVATION

Dharohar 4D: Bridging Time, Geometry & Authenticity

A holistic, end-to-end platform turning scattered media into interactive, verified 4D twins.

SPATIAL + TEMPORAL

1. 4D Digital Twins

- 3D photogrammetric geometry combined with the 4th dimension (Time).
- Explore monuments across historical eras: 11th-century Solanki, 1890 ASI colonial surveys, and 2026 current state.
- Interactive solar simulation matching monument alignment.

100% VERIFIABLE

2. Evidence Grounding

- Dual-stage verification pipeline with zero tolerance for AI hallucination.
- Every architectural claim is cross-verified against primary literature.
- Fuzzy matching guarantees claims are backed by verbatim citations.

ACCESSIBLE TO ALL

3. Mobile WebXR AR

- No app installation required; runs natively on iOS & Android browsers.
- Project ancient stepwells and temple sanctums into physical reality at 1:1 scale.
- Multilingual audio guide in English, Hindi, and Gujarati.

INSTITUTIONAL GRADE

4. Cryptographic Trust

- SHA-256 asset finger-printing prevents tampering or metadata alteration.
- Curatorial audit trail with side-by-side archival diff review.
- Exportable ASI Verification Certificates with cryptographic hashes.

ENGINEERING DESIGN

4-Tier Production Architecture

Designed for high throughput, cryptographic provenance, and instant mobile responsiveness.

INGESTION & STORAGE

Layer 1: Multi-Modal Ingestion & Archive

- Automated EXIF extraction (focal length, GPS, timestamp).
- Perceptual image hashing (pHash) for deduplication & tamper detection.
- SQLite FTS5 full-text search with multilingual query expansion.

VISUAL & GEOMETRY

Layer 2: 3D Photogrammetry & 4D Temporal Engine

- Multi-view photogrammetry reconstruction generating optimized GLB/glTF twins.
- Dynamic solar azimuth calculation for Solanki sun-temple alignments.
- Stepwell water level simulation & aging material shaders.

RIGOR & TRUST

Layer 3: Authenticity & Claim Verification Engine

- LLM extraction strictly constrained to uploaded archival documents.
- RapidFuzz fuzzy verification verifying exact text snippets.
- Cryptographic SHA-256 certificate generation for verified assets.

USER EXPERIENCE

Layer 4: Immersive Frontend & Reviewer Portal

- Next.js 14 App Router with React Three Fiber & Google Model-Viewer.
- Interactive India Geo-Explorer map with coordinate pins & dynasty filters.
- Side-by-side 3D review diff viewer with 1-click evaluator bypass.

USER EXPERIENCE

Interactive 3D Digital Twin & Mobile WebXR AR

Bringing cultural heritage into the physical world through cutting-edge browser capabilities.

SPATIAL COMPUTING

WebXR AR Experience

- Instant 1-click AR mode via WebXR & iOS Quick-Look.
- Zero external app downloads required.
- Place full-scale 3D monuments on tables, floors, or open grounds.
- Real-world scale perception for architectural education.

SCHOLARLY ANNOTATION

Curated 3D Hotspots

- Interactive 3D pins placed directly on architectural features.
- Click to inspect: Solanki pillar carvings, octagonal well shafts, torana gates.
- Every hotspot displays verified primary literature citations & page numbers.
- Direct connection between 3D geometry and historical scholarship.

MULTI-SENSORY

Solar Lighting & Audio Tour

- Simulate Sunrise, Golden Hour, and Night illumination in real-time.
- Observe solar alignment of temples (e.g. Modhera equinox rays).
- Integrated Web Speech TTS narration.
- Listen to architectural explanations in English, Hindi, and Gujarati.

EXPLORATION TOOLS

Spatiotemporal Heritage: Geo-Explorer & Time-Slider

Navigate India's sacred geography and witness architectural transformation across centuries.

GEOGRAPHIC DIMENSION

India Geo-Explorer Interactive Map

- Cartographic visualization mapping protected heritage sites across India.
- Interactive coordinate pins with instant 3D twin previews.
- Filter monuments by Dynasty (Solanki, Vaghela, Eastern Ganga, Chola).
- State-level architectural clustering (Gujarat, Odisha, Tamil Nadu).
- Bridges regional heritage data into one unified national map.

TEMPORAL DIMENSION (4D)

Then-vs-Now Archival Comparison Slider

- Interactive split-screen comparison slider.
- Side A: 1890 James Burgess ASI colonial photographs & archaeological plates.
- Side B: 2026 photogrammetric 3D digital twin of the current monument state.
- Examine structural weathering, missing sculptures, and ASI conservation work.
- Essential tool for archaeological damage assessment and educational walkthroughs.

AUTHENTICITY ENGINE

Curatorial Rigor & Cryptographic Verification

Preventing historical falsification through cryptographic provenance and peer review.

DATA INTEGRITY

Cryptographic Hashing

- Every uploaded photo, 3D model, and archival text receives a SHA-256 fingerprint.
- Immutable digital ledger guarantees no unrecorded alterations occur.
- Perceptual hashing (pHash) detects cropped, resized, or re-compressed duplicates.

ANTI-HALLUCINATION

Automated Claim Verification

- LLM extracts claims with mandatory snippet citations.
- Fuzzy string matcher verifies snippet exists verbatim in source PDF/doc.
- Unverified or contradictory claims are rejected automatically with zero human bias.

INSTITUTIONAL VALUE

ASI Verification Certificates

- Generates official digital Authenticity Certificates.
- Contains monument metadata, epoch timestamps, curator signature, and SHA-256 seal.
- Exportable as JSON metadata and printable formal verification documents.

EVALUATOR EXPERIENCE

Modern Reviewer Portal & Side-by-Side Review Diff

Streamlined workflow empowering ASI curators and competition evaluators.

FRICTIONLESS ACCESS

1-Click Evaluator Access

- No credentials barrier for hackathon judges: 1-click evaluator bypass button.
- Optional Magic Link token authentication for secure administrative sessions.
- Role-based access control preventing unauthorized state modifications.

QUALITY ASSURANCE

Side-by-Side Review Diff

- Compare pending digital twin submissions directly against approved reference models.
- Inspect texture resolution, dense point count, and reprojection error metrics.
- Curator can approve, flag for re-capture, or reject with architectural feedback.

100% RELIABILITY

Resilience & Offline Fallback

- Includes complete offline showcase dataset (Adalaj, Modhera, Rani Ki Vav, Konark).
- Auto-switching interceptor detects network dropouts seamlessly.
- Gold shimmering skeleton loaders provide a polished 60 FPS experience.

DEPLOYMENT & DEVOPS

Full-Stack Technology Stack & Production Cloud Deployment

Built on battle-tested open technologies and deployed live on Render.

CLIENT LAYER

Modern Frontend

- Next.js 14 (App Router) for high-performance server-side rendering.
- React Three Fiber & Three.js for interactive 3D rendering.
- Google Model-Viewer for standard-compliant WebXR AR.
- Tailwind CSS for responsive heritage-themed dark-gold UI.

API & ENGINE

High-Speed Backend

- FastAPI (Python 3.11) with asynchronous endpoints.
- SQLite + FTS5 for lightning-fast full-text search across documents.
- RapidFuzz for high-speed fuzzy text citation verification.
- Groq Llama 3.3 for grounded claim extraction.

PRODUCTION CLOUD

Cloud Infrastructure

- Render Cloud Blueprint (render.yaml) for automated Infrastructure-as-Code.
- Dual Web Services: Python FastAPI Backend + Next.js Frontend.
- Automatic zero-downtime CI/CD sync from GitHub main branch.
- Live health-checks on /healthz endpoint.

CONCLUSION & IMPACT

Preserving the Past, Empowering the Future

Dharohar 4D solves the critical tension between cutting-edge AI immersion and scholarly authenticity.

✓ Overcomes physical loss by archiving multi-epoch architectural baselines.
✓ Delivers zero-install WebXR AR directly in mobile browsers.
✓ Provides 100% citation-grounded authenticity with cryptographic certificates.
✓ Production-deployed, live on Render, and ready for institutional ASI adoption.

Live Web App: https://dharohar-4d.onrender.com • Thank You!